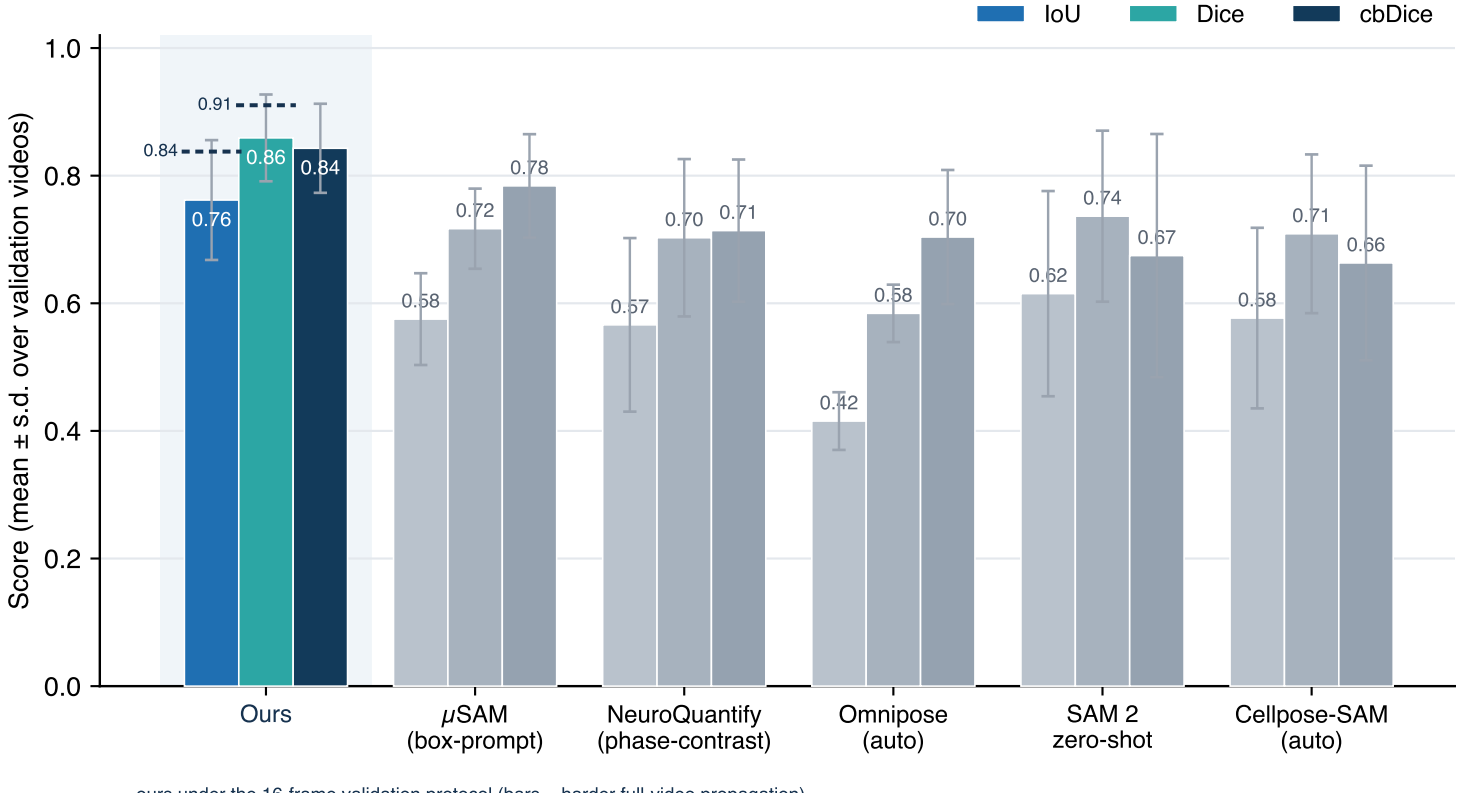
Neuron segmentation on the validation set — ours leads every metric, from a single frame-0 annotation
IoU
Dice
cbDice
Score (mean ± s.d. over validation videos)
1.0
0.8
0.6
0.4
0.2
0.0
0.91
0.84
0.86
0.84
0.76
0.58
0.72
0.78
0.57
0.70
0.71
0.42
0.58
0.70
0.62
0.74
0.67
0.58
0.71
0.66
Ours
μSAM (box-prompt)
NeuroQuantify (phase-contrast)
Omnipose (auto)
SAM 2 zero-shot
Cellpose-SAM (auto)
ours under the 16-frame validation protocol (bars = harder full-video propagation)